

TECHNICAL DOSSIER

SCARA

SCARA robot / industrial robotic arm intended to become an autonomous chess-playing robot.

DOSSIER DATA

PROJECT LEAD

Justo Darío Valverde Torres and Marcos Moreno

STATUS

In development

REPOSITORY

github.com/RoboTech-URJC/Robot-Scara

LAST UPDATE

June 2026

Project Description

The Scara project began at the end of the previous academic year and has continued to be developed during the current one.

The Scara project consists of developing a SCARA robot / industrial robotic arm. In addition to the large amount of knowledge gained during its development, the goal is to turn the Scara arm into an autonomous chess-playing robot capable of detecting when a person is sitting in front of it and moving the chess pieces placed before it according to the game situation.

Justo Darío Valverde Torres and Marcos Moreno are the project leads for this project.



Work Completed During the Year

Phase 1: at the beginning of the year, the team resumed assembly work on the hardware that had not been completed the previous year.

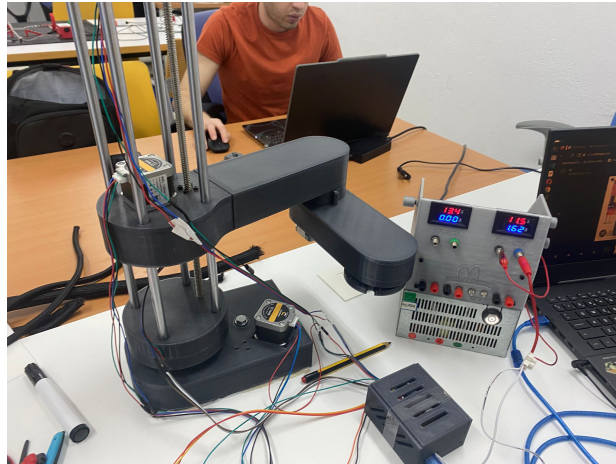
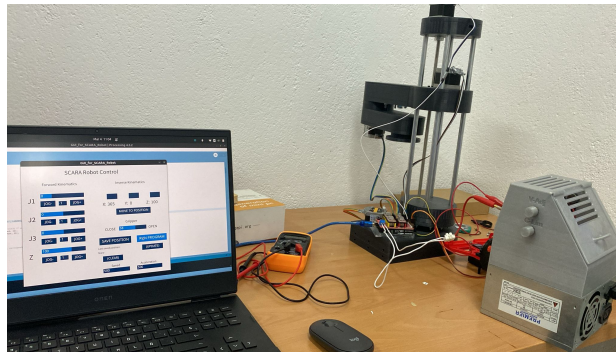
The Scara robot was made possible thanks to a donation to the association. This donation consisted of a partially assembled robotic arm. The 3D prints, belt systems, and motors were assembled, but it had no electronics or wiring.

This phase, which began last year, consisted of becoming familiar with the Scara hardware and designing how to correctly connect the components in order to bring the robotic arm to life.



Phase 2: once the Scara was almost fully assembled, before designing and mounting an end-effector, the mechanical and electrical system design of the robot was validated.

Using an Arduino UNO, movement loops were programmed to help verify that the hardware was working correctly. Later, the first version of the gripper was designed, but it turned out to be too large for the goal of making Scara play board games.



Phase 3: once it had been verified that the robot worked well mechanically, the team moved into developing a demo in which Scara would play tic-tac-toe.

Still using the Arduino UNO as the hardware controller, connected to a laptop through serial communication, the system would run computer vision to detect the current board state, a tic-tac-toe algorithm, and a user interface for program debugging and graphical support.

In this way, the computer detects the opponent's moves and chooses the move that it will communicate to the Arduino, which in turn commands the motors to translate that move into reality.



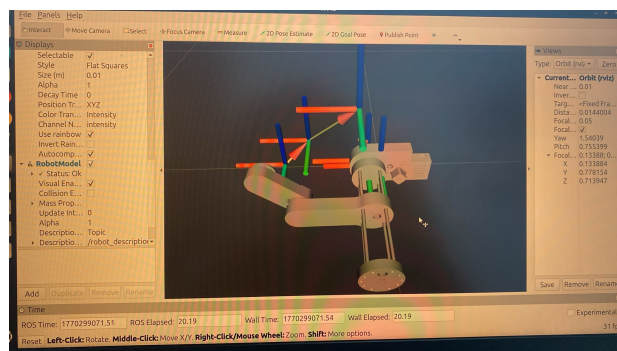
Current Phase

The following tasks are currently being developed in parallel:

- **End-effector design:** the design and 3D printing of two prototype end-effector models is underway. One is actuated by an electromagnet so that it can easily pick up magnetized or metallic chess pieces. The other has a more general purpose and consists of a two-finger gripper actuated by a servomotor.



- **Software migration and development board change to an SBC:** in order to meet industry standards and the requirements of the chess-playing robot application, the current development board, an Arduino UNO, will be replaced by a Jetson Orin Nano single-board computer.



The Jetson Orin Nano will be able to run artificial intelligence models and work across several threads performing multiple tasks, in addition to hardware control, without needing external computing power.

This change also involves migrating the software to ROS2. A simulation environment is currently being prepared to validate the algorithms and the system once completed. To do this, work has begun by creating the robot's URDF model and becoming familiar with ROS2 MoveIt.

Next Goals

- Finish developing the Scara simulation.
- Finish designing the end-effectors and test their effectiveness.
- Test different approaches for robot control: ROS2 MoveIt and reinforcement learning in Isaac Gym.
- Develop computer vision nodes in ROS2 for both tic-tac-toe and chess.
- Develop or implement a chess algorithm as a ROS2 node.
- Develop hardware interfaces for Nema17 stepper motors, which will be needed once MoveIt is tested on the real robot.
- Once the previous goals have been validated in simulation, update the hardware to the Jetson Orin Nano and test the behaviors on the real robot.